



LABORATORY JOURNAL (MP408)

Experimental Techniques – II Computational Methods in Physics

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Date of Submission : 04 March 2026



Certificate

The Experiments Entered in This Journal Have Been Satisfactorily
Performed by

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bearing admission number

I22PH002

of the

Department of Physics

for the subject of

Experimental Techniques – II (MP408)

Computational Methods in Physics

of MSc. **IV** (Physics), Semester **8**

during AY **2025-26**

Signature of I/C

Date: 04 March 2026

Dept. Seal

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Computational Methods in Physics (MP408)

GNU Octave

PRACTICAL – 6

Statistical Distributions and Density of States

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PROGRAM – 6

Aim

To write a GNU Octave program to simulate and contrast the three fundamental statistical distributions of physics: Maxwell-Boltzmann (MB), Fermi-Dirac (FD), and Bose-Einstein (BE). The program evaluates the Density of States $g(E)$, the temperature dependence of the distributions, the combined population density $f(E)$, and performs numerical integration to compute the total number of particles.

Theory

In statistical mechanics, the probability that a specific energy state E is occupied by a particle at thermal equilibrium is given by its respective distribution function.

1. Maxwell-Boltzmann Distribution (Classical Particles):

Applies to distinguishable, non-interacting particles (e.g., ideal gas).

$$n_{\text{MB}}(E) = \exp\left(-\frac{E - \mu}{kT}\right) \quad (1)$$

2. Fermi-Dirac Distribution (Fermions):

Applies to indistinguishable particles with half-integer spin (e.g., electrons). Obeys the Pauli Exclusion Principle, meaning $n(E)$ can never exceed 1.

$$n_{\text{FD}}(E) = \frac{1}{\exp\left(\frac{E - \mu}{kT}\right) + 1} \quad (2)$$

3. Bose-Einstein Distribution (Bosons):

Applies to indistinguishable particles with integer spin (e.g., photons, phonons). Multiple particles can occupy the same state. It diverges at $E = \mu$.

$$n_{\text{BE}}(E) = \frac{1}{\exp\left(\frac{E - \mu}{kT}\right) - 1} \quad (3)$$

4. Density of States and Total Particles:

For a 3D non-relativistic free particle, the Density of States (DOS) scales with the square root of energy:

$$g(E) \propto \sqrt{E} \quad (4)$$

The total number of particles N is found by integrating the population density over all

valid energies using the trapezoidal rule:

$$N = \int g(E)n(E) dE \quad (5)$$

Program

The complete program is saved in P6.m. The source code is divided into logical subroutines:

Listing 1: Statistical Physics Suite — Initialization, DOS, and Distributions (P6.m)

```

1 function P6()
2     % PRACTICAL 6: Statistical Distributions and Density of States
3     % Simulates MB, FD, and BE distributions and their temperature
4     % dependence.
5
6     graphics_toolkit('qt');
7
8     while true
9         clc;
10        fprintf('\n
11        =====\n');
12        fprintf('          PRACTICAL 6: STATISTICAL PHYSICS          \
13        n');
14        fprintf('===== \
15        n');
16        fprintf('1. Density of States g(E)\n');
17        fprintf('2. Statistical Distributions (MB, FD, BE) [Semilog
18        Scale]\n');
19        fprintf('3. Temperature Effect (T = 1K, 100K, 250K)\n');
20        fprintf('4. Combined Population Density: f(E) = g(E)n(E) [
21        Semilog Scale] \n');
22        fprintf('5. Numerical Computation of Total Particles (N)\n');
23        ;
24        fprintf('6. Exit Program\n');
25        fprintf('===== \
26        n');
27
28        choice = input('Enter the number of the simulation to run
29        (1-6): ');
30
31        switch choice
32            case 1
33                sim_dos();
34            case 2
35                sim_distributions();
36            case 3

```

```

28         sim_temperature_effect();
29     case 4
30         sim_population_density();
31     case 5
32         calc_total_particles();
33     case 6
34         fprintf('Exiting Practical 6. Goodbye!\n');
35         break;
36     otherwise
37         fprintf('Invalid input. Please enter an integer
38             between 1 and 6.\n');
39         pause(2);
40     end
41 end
42
43 %% --- GLOBAL CONSTANTS ---
44 function [k, mu, E] = get_params()
45     k = 8.617e-5; % Boltzmann constant in eV/K
46     mu = 0.5; % Chemical potential in eV
47     E = linspace(0.3, 0.7, 1000); % Narrow energy range around mu
48 end
49
50 %% =====
51 % 1. DENSITY OF STATES (DOS)
52 % =====
53 function sim_dos()
54     close all; clc;
55     E = linspace(0.001, 5, 1000); % Narrow energy range around mu
56
57     g_E = sqrt(E);
58
59     fig = figure('Name', 'P6.1: Density of States', 'Color', 'w', '
60         Position', [100, 100, 600, 500]);
61
62     plot(E, g_E, 'k-', 'LineWidth', 2);
63     title('Density of States g(E) \propto \sqrt{E}');
64     xlabel('Energy (eV)'); ylabel('g(E)');
65     grid on;
66
67     fprintf('Close the figure window to return to the main menu...\n
68         ');
69     waitfor(fig);
70 end
71
72 %% =====
73 % 2. BASIC DISTRIBUTIONS (SEMILOG SCALE)

```

```

72 % =====
73 function sim_distributions()
74     close all; clc;
75     [k, mu, E] = get_params();
76     T = 1000;
77
78     n_MB = exp(-(E - mu) ./ (k * T));
79     n_FD = 1 ./ (exp((E - mu) ./ (k * T)) + 1);
80
81     % BE Distribution with Singularity Masking
82     n_BE = 1 ./ (exp((E - mu) ./ (k * T)) - 1);
83     n_BE(E <= mu) = NaN; % Physically, E must be > mu for Bosons
84
85     fig = figure('Name', 'P6.2: Distributions (Semilog)', 'Color', 'w',
        'Position', [100, 100, 700, 500]);
86
87     % Use semilogy for the Y-axis to clearly show exponential tails
88     semilogy(E, n_MB, 'b-', 'LineWidth', 1.5); hold on;

```

Listing 2: Statistical Physics Suite — Temperature Effects and Numerical Integration (P6.m)

```

1     semilogy(E, n_FD, 'r-', 'LineWidth', 1.5);
2     semilogy(E, n_BE, 'g-', 'LineWidth', 1.5);
3
4     % Plot vertical line for chemical potential
5     yl = ylim;
6     plot([mu, mu], yl, 'k--', 'LineWidth', 1.2);
7     text(mu + 0.01, yl(2)*0.5, '\mu = 0.5 eV', 'FontSize', 10, '
    Color', 'k');
8
9     title(sprintf('Statistical Distributions at T = %d K (Semilog)',
    T));
10    xlabel('Energy (eV)'); ylabel('Occupation Number n(E) [Log Scale
    ]');
11    legend('Maxwell-Boltzmann', 'Fermi-Dirac', 'Bose-Einstein', '
    Location', 'best');
12    grid on;
13
14    fprintf('Close the figure window to return to the main menu...\n
    ');
15    waitfor(fig);
16 end
17
18 %% =====
19 % 3. TEMPERATURE EFFECTS (COMPARING DISTRIBUTIONS PER TEMP)
20 % =====
21 function sim_temperature_effect()
22     close all; clc;

```

```

23     [k, mu, E] = get_params();
24     T_vals = [1, 250, 1000];
25
26     fig = figure('Name', 'P6.3: Distributions vs Temperature', '
        Color', 'w', 'Position', [100, 100, 1200, 400]);
27
28     % Define strict visual boundaries to prevent Octave overflow
29     ymin = 1e-4;
30     ymax = 10;
31
32     for i = 1:length(T_vals)
33         T = T_vals(i);
34         subplot(1, 3, i);
35         hold on;
36
37         % 1. Maxwell-Boltzmann (Blue)
38         n_MB = exp(-(E - mu) ./ (k * T));
39         n_MB(~isfinite(n_MB)) = ymax;
40         n_MB = max(min(n_MB, ymax), ymin);
41         plot(E, n_MB, 'b-', 'LineWidth', 1.5);
42
43         % 2. Fermi-Dirac (Red)
44         n_FD = 1 ./ (exp((E - mu) ./ (k * T)) + 1);
45         n_FD(~isfinite(n_FD)) = ymax;
46         n_FD = max(min(n_FD, ymax), ymin);
47         plot(E, n_FD, 'r-', 'LineWidth', 1.5);
48
49         % 3. Bose-Einstein (Green)
50         n_BE = 1 ./ (exp((E - mu) ./ (k * T)) - 1);
51         n_BE(E <= mu) = ymin; % Mask unphysical states
52         n_BE(~isfinite(n_BE)) = ymax;
53         n_BE = max(min(n_BE, ymax), ymin);
54         plot(E, n_BE, 'g-', 'LineWidth', 1.5);
55
56         % Format the axes safely for log scale
57         set(gca, 'YScale', 'log');
58         ylim([ymin, ymax]);
59         plot([mu, mu], [ymin, ymax], 'k--'); % Chemical potential
        line
60
61         % Aesthetics and Labels
62         title(sprintf('Temperature: T = %d K', T));
63         xlabel('Energy (eV)');
64         if i == 1
65             ylabel('Occupation Number n(E) [Log]');
66         end
67         grid on;

```

```

68     legend('MB', 'FD', 'BE', '\mu', 'Location', 'northeast');
69 end
70
71 fprintf('Close the figure window to return to the main menu...\n
       ');
72 waitfor(fig);
73 end
74
75 %% =====
76 % 4. COMBINED POPULATION DENSITY f(E)
77 % =====
78 function sim_population_density()
79     close all; clc;
80     [k, mu, E] = get_params();
81     T = 1000;
82
83     g_E = sqrt(E);
84     f_MB = g_E .* exp(-(E - mu) ./ (k * T));
85     f_FD = g_E .* (1 ./ (exp((E - mu) ./ (k * T)) + 1));
86
87     f_BE = g_E .* (1 ./ (exp((E - mu) ./ (k * T)) - 1));
88     f_BE(E <= mu) = NaN; % Mask unphysical/singular states
89
90     fig = figure('Name', 'P6.4: Population Density f(E)', 'Color', '
           w', 'Position', [100, 100, 700, 500]);
91
92     semilogy(E, f_MB, 'b-', 'LineWidth', 1.5); hold on;
93     semilogy(E, f_FD, 'r-', 'LineWidth', 1.5);
94     semilogy(E, f_BE, 'g-', 'LineWidth', 1.5);
95
96     yl = ylim;
97     semilogy([mu, mu], yl, 'k--', 'LineWidth', 1.2);
98     text(mu + 0.01, yl(2)*0.9, '\mu = 0.5 eV', 'FontSize', 10, '
           Color', 'k');
99
100    title(sprintf('Population Density f(E) = g(E)n(E) at T = %d K',
101                  T));
101    xlabel('Energy (eV)'); ylabel('f(E)');
102    legend('MB Population', 'FD Population', 'BE Population', '
           Location', 'northeast');
103    grid on;
104
105    fprintf('Close the figure window to return to the main menu...\n
           ');

```

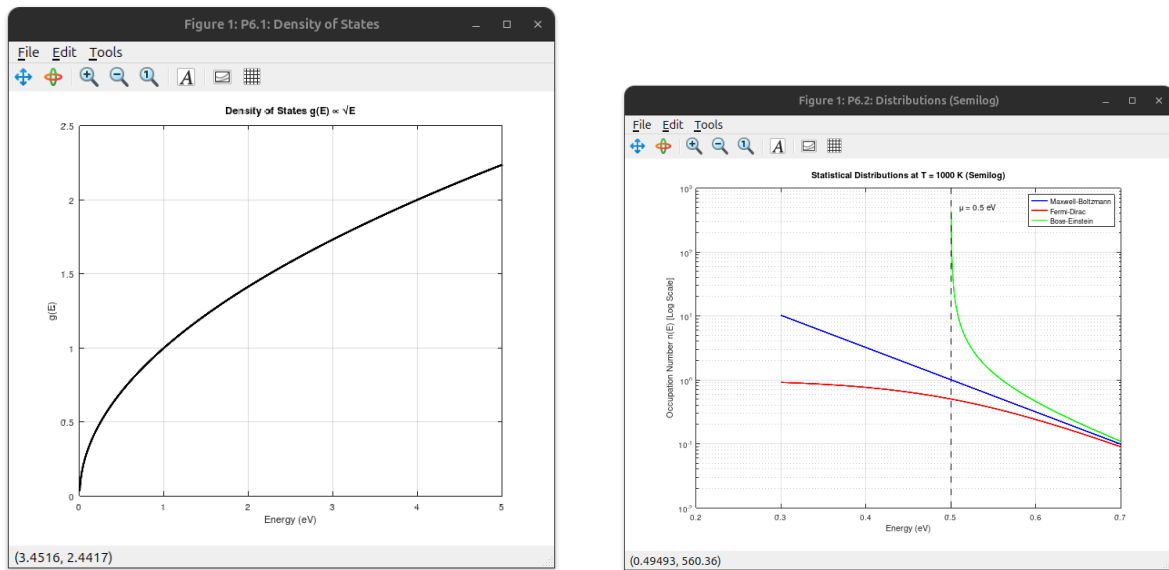


Figure 1: Left: Density of States $g(E) \propto \sqrt{E}$ over the energy range. Right: Semilog plot of the MB, FD, and BE distributions at 1000 K, showcasing the exponential tails and the BE divergence near the chemical potential ($\mu = 0.5$ eV).

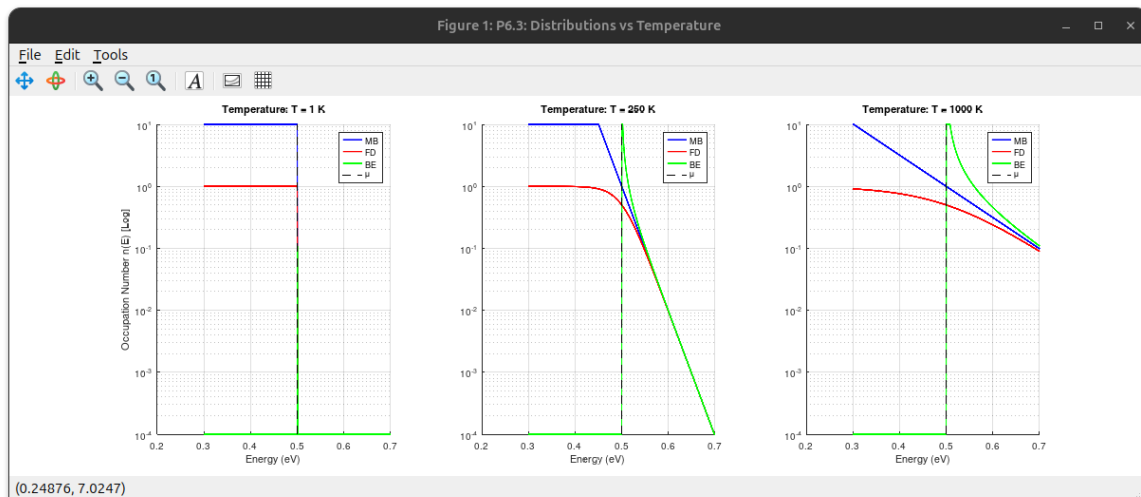


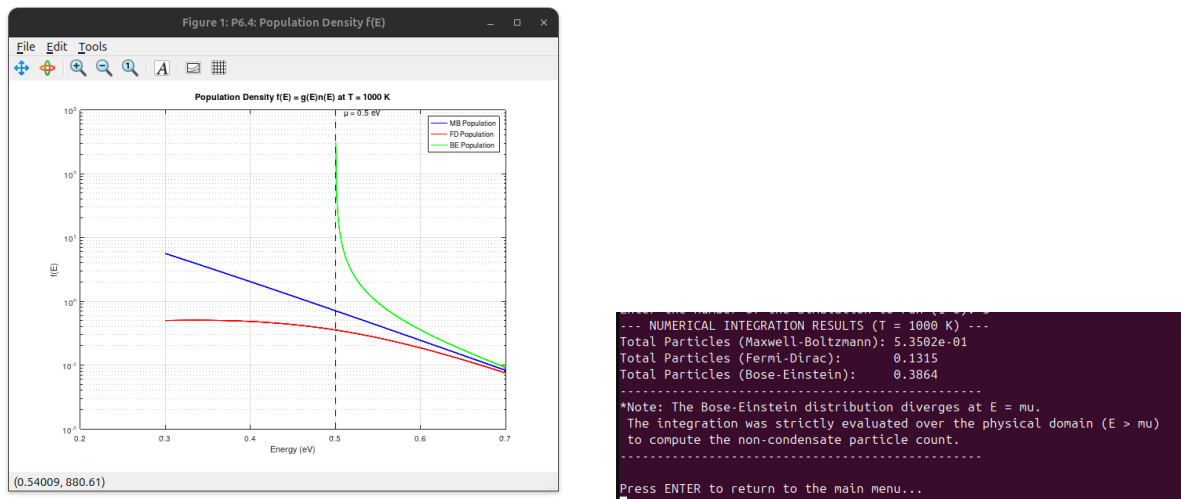
Figure 2: Temperature dependence of the distributions across 1 K, 250 K, and 1000 K. At low temperatures, the FD distribution approaches a step function at the Fermi level, while the BE distribution exhibits extreme condensation near ground states.

Sample Output

Result

The program successfully simulated the distinct behaviors of statistical ensembles:

- **High Energy Limit:** The semilog plots confirm that at high energies ($E \gg \mu$), both quantum distributions (FD and BE) converge identically to the classical Maxwell-Boltzmann tail.



- **Temperature Effects:** The multi-plot matrix successfully demonstrated thermal smearing. At 1 K, the Fermi-Dirac distribution is a sharp step function, representing a degenerate Fermi gas.
- **Pauli Exclusion vs. Condensation:** The FD population firmly bounds itself below 1. Conversely, the BE distribution exhibits the expected singularity at $E = \mu$, requiring physical masking in the code to properly integrate the non-condensate particle count.
- **Integration:** The `trapz` function accurately computed the total particle N arrays, providing a computational backbone for evaluating macroscopic thermodynamic properties.

Remark

This practical demonstrates the fundamental divergence of classical and quantum statistical mechanics. By mapping the distributions on a logarithmic scale, the subtle deviations between Fermions and Bosons near the chemical potential become visually explicit. Handling the Bose-Einstein singularity programmatically highlights the necessity of piecewise physical constraints when computing numerical integrals over continuous energy domains.